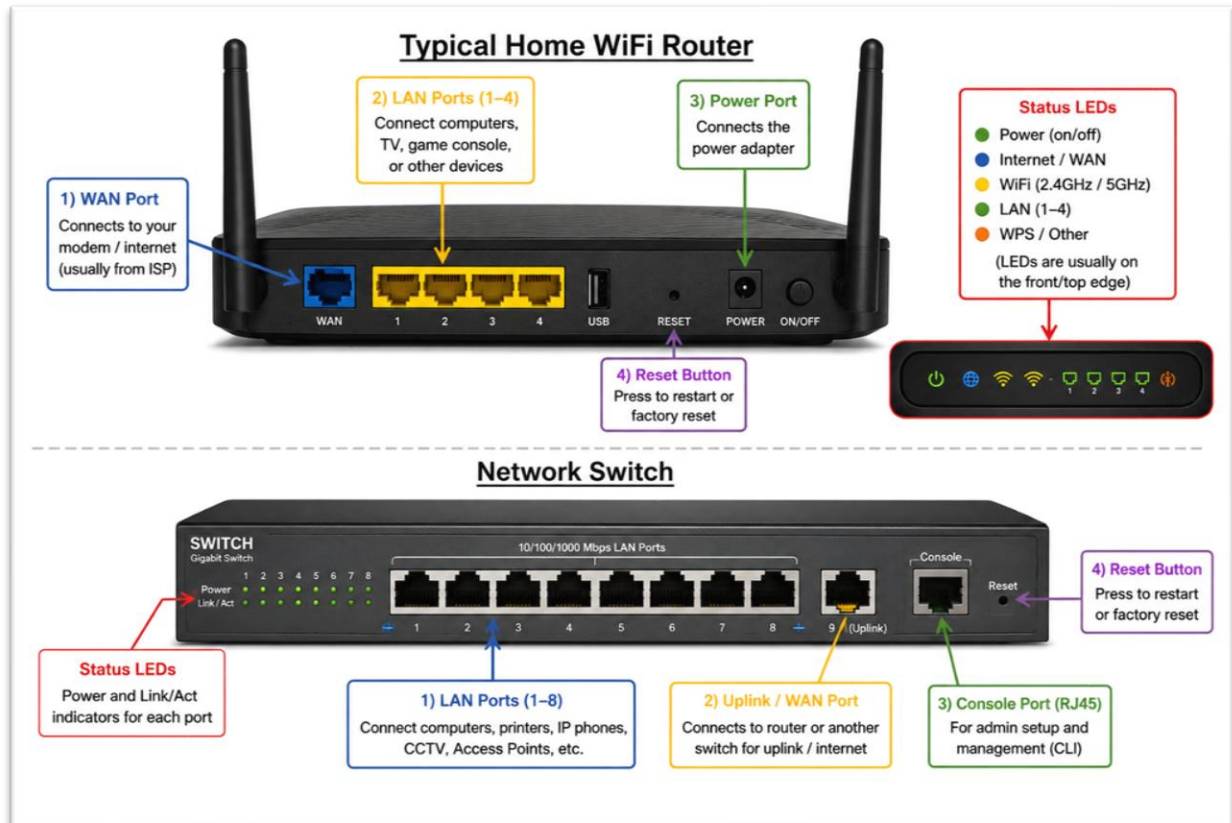


Module-3 (Assignment Day-2)

1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

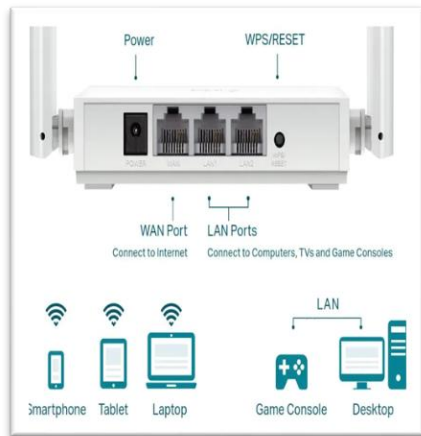


Point	Router	Switch
Main function	Connects different networks, such as a home network to the Internet.	Connects multiple devices on the same local network (LAN).
Address used	Uses IP addresses to identify and route data between networks.	Uses MAC addresses to identify and forward data to devices.
Traffic handling	Sends packets from one network to another and selects the appropriate route.	Forwards frames to the correct device within the same network, reducing unnecessary traffic.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



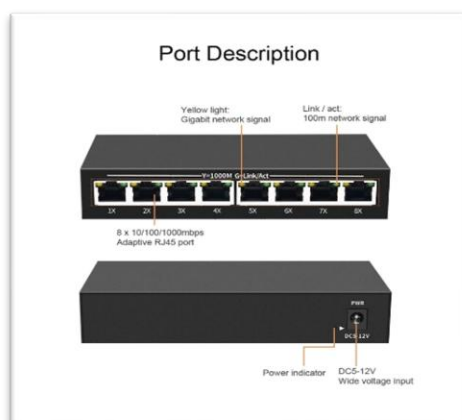
1. Wi-Fi Router:



Identification:

- **Power port:** Usually at the back; connects the router to its power adapter.
- **Ethernet/LAN Port 1:** Used to connect a computer or other wired device.
- **Ethernet/LAN Port 2:** Used to connect another wired device.
- **WAN/Internet port:** Connects the router to the modem/Internet connection.
- **LED indicators:** Usually on the front; show **Power, Internet, Wi-Fi, and LAN** activity.

2. Network Switch:



Identification:

- **Power port:** Usually on the back or side; connects the switch to its power adapter.
- **Ethernet Port 1:** Connects a computer or another network device.
- **Ethernet Port 2:** Connects another computer or network device.
- **Other Ethernet ports:** Used to connect additional devices to the same network.
- **LED indicators:** Usually near the Ethernet ports; show **power, connection, and network activity**.

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.



Small Gaming Cafe Network Scenario

I am setting up a small gaming cafe with **8 gaming PCs** for IPL streaming and multiplayer games.

1. First, I would connect the **ISP Internet line to the router's WAN/Internet port**. The router provides Internet access and gives IP addresses to the devices in the café.
2. Then, I would connect one **LAN port of the router to a network switch**.
3. I would connect the **8 gaming PCs to 8 Ethernet ports on the switch** using Ethernet cables.
4. All 8 PCs can then communicate with each other through the switch and access the Internet through the router.

Why this setup:

The **router connects the cafe network to the ISP/Internet**, while the **switch provides enough Ethernet ports for all 8 PCs** and efficiently handles traffic between them. This setup is simple, fast, and suitable for multiplayer gaming and IPL streaming.